

# CSCI 447: Machine Learning – Fall 2026

Class meeting time (Fall): MWF, 1:10 – 2:00, GH 100

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Office hours: MWF, 2:10 – 3:00 or by appointment

Textbook: Ethem Alpaydin, *Introduction to Machine Learning*, 4th edition, Cambridge, MA: The MIT Press, 2020. Additional select readings as appropriate.

## Course Outline

This course is an advanced undergraduate/introductory graduate course that introduces students to the fundamental methods in machine learning as well as to several biologically-inspired approaches to developing advanced, artificially intelligent systems using principles from biology and machine learning. The major topics to be covered include linear models, logic-based models, non-parametric methods, unsupervised learning, neural networks, evolutionary algorithms (e.g., genetic algorithms, genetic programming, and differential evolution), and swarm intelligence (e.g., particle swarm optimization and ant colony optimization). Students will be introduced to the fundamental algorithms in each of these topics. The primary goals of this course include providing the student with the skills to assess methods of machine learning and to build systems incorporating various machine learning techniques.

The four major topics expected to be covered include the following:

- Fundamentals of Machine Learning
- Principal Machine Learning Algorithms
- Neural Networks
- Population-Based Algorithms

## Learning Outcomes

At the end of this course, the student will be able to:

- Formulate and assess problems in machine learning.
- Understand and be able to implement several machine learning algorithms.
- Assess the strengths and weaknesses of alternative representations and algorithms in machine learning.
- Analyze the expected performance of machine learning algorithms based on formal principles from the analysis of algorithms.
- Assess the performance of machine learning algorithms based on principles of sound experimental design and empirical analysis.

## Course Evaluation

The best way to learn about techniques in machine learning is by implementing approaches to solve different problems. Therefore, the assignments in the course will emphasize implementation and experimentation. Each programming project will involve programming specific machine learning algorithms and testing the algorithms on multiple data sets. Four projects will be assigned, and each project will carry with it the requirement to write a short expository paper describing the results of the associated experiments. Note

that *algorithm* libraries (e.g., TensorFlow, PyTorch, scikit-learn, weka) are not permitted to be used to implement any of the algorithms. Using libraries such as NumPy and Pandas is permitted to implement *basic* numerical and data structure functions. Note also that ChatBot and other Generative AI-based tools are expressly prohibited from being used on all aspects of the course.

In addition to the programming assignments, as a way to evaluate what has been learned, as opposed to what online tools know, the practice of administering closed-book examinations is being reinstituted. A midterm exam and a final exam will be administered. No aids will be permitted during the exams, including cell phones, calculators, tablets, computers, or cheat sheets. The exams will consist of three parts: 1) objective-style questions, 2) specific questions on the programming assignments, and 3) open-ended short-answer questions.

Grading requirements and due dates are summarized as follows:

- Project 1 (Assigned August 26, 2026; Due: September 18, 2026): 15%
- Project 2 (Assigned September 21, 2026; Due October 14, 2026): 15%
- Midterm Examination (October 16, 2026): 15%
- Project 3 (Assigned October 19, 2026; Due November 9, 2026): 15%
- Project 4 (Assigned November 11, 2026, Due December 11, 2026): 15%
- Final Examination (December 14, 2026, 12:00 pm): 25%

All assignments are due by 11:59 pm on the assigned date. Assignments must be uploaded into Canvas using the group submission feature. Only one person on the team needs to upload the submission. Written documents shall be PDF and shall conform to the JMLR format (<http://www.jmlr.org/format/format.html>). Only project reports (no code) will be submitted.

All project reports must be no more than 15 pages using the JMLR format and include the following:

- Title and authors
- Problem Statement with Testable Hypothesis
- Experimental Approach and explanation of project design
- Explanation of the algorithms and how they were implemented (without showing code)
- Experimental Results (tables and/or graphs with explanation)
- Analysis and Discussion of Results
- Conclusion
- References (if using any not provided in course content)
- Two appendices (not included in page limit):
  - Main lessons learned
  - Delineation of who did what on the project

**Group Work:** The four projects will be completed in groups of two people. On occasion, due to course drops, it may become necessary to create groups of three. Under no circumstances will students be permitted to complete the projects on their own. The groups will be assigned during the first week of class. It is up to the individual groups to ensure fair distribution of labor. All members of the group will receive the same grade. On all projects, any programming language may be used, but all group members must be familiar with and will use the same language. Note that all members of the team must code their share

(1/2 or 1/3, depending on group size) of the project. During the first assignment, course withdrawals will not be considered after the self-withdrawal date. For all other assignments, course withdrawals will only be considered during the first five days of the assignment to ensure a team is not left hanging due to a drop.

**Appeals:** With respect to grading appeals, ideally, assignments will be returned within one week of being handed in. Questions on grading should be directed to the TA or grader, assuming one is assigned, within one week of receiving the graded assignment. After receiving a decision from the TA, additional appeals may be made to the instructor within one week of receiving that decision. After that, appeals will not be considered. The instructor's decision is final. If the course has no TA or grader, then appeals must be made directly to the instructor within one week of receiving the graded assignment.

**Late Submissions:** In general, all assignments are expected to be turned in on time. The default policy is that late submissions will receive no credit. That said, issues that are illness-related or due to an unexpected emergency will be considered on a case-by-case basis. Notification must be given to the instructor as soon as possible for such situations affecting any group. Issues related to time management or personal (non-school) commitments will not be considered.

## Academic Conduct and Academic Integrity

All students are expected to conduct themselves in a professional, honorable, and courteous way. Plagiarism and other forms of academic misconduct, including seeking out solutions to homework assignments or unauthorized collaboration, will have severe consequences. *Example: Using StackExchange or LLMs to get coding help and then submitting the code as your own.* Students who engage in any form of academic misconduct will be reported to the Dean of Students and will receive an F for the course. Dropping the course to avoid an F resulting from academic misconduct will not be permitted under any circumstances.

## Policy on the Use of Generative AI/Large Language Models

This class will strive to create an environment that fosters learning, critical thinking, effective communication, and technology development. To achieve these goals, using generative AI-based tools such as ChatGPT, Claude, Co-Pilot, or similar are prohibited in this course.

While ChatGPT, Claude, and other large language models can be powerful and useful tools in certain contexts, relying on them for this course undermines the learning objectives. Students are being taught to understand the fundamentals of machine learning, rather than to become a *user* of ML or AI tools. This approach involves developing skills in independent thinking, problem-solving, and engagement with the subject matter. By restricting the use of large language models, knowledge, creativity, and critical analysis will be used to complete all assignments and actively participate in class discussions.

Advanced, critical thinking also requires the development of strong communication skills. This includes engaging in conversation and discussions, as well as writing. GenAI cannot be used in any of the writing assignments either, except when doing spell checking and basic grammar checking. Under no circumstances may GenAI tools be used for composition, summarization, or style improvement. Note that TurnItIn generates an "AI score," and all high AI scores will be investigated.

It is important to emphasize that this requirement applies to all aspects of the course, including programming assignments, writing assignments, and any form of communication related to the course content. Use of any generative AI tools during these activities will be considered a violation in the student code of conduct and will be reported as academic misconduct to the Dean of Students.

## Class Topics (subject to revision): Check schedule above for due dates

- Week 1 (August 26, 28)
  - *Topic:* Introduction to Machine Learning (Learning Paradigms, First Algorithm, Experimental Methods)
  - *Readings:* Alpaydin, Chapter 1, 20
- Week 2 (August 31, September 2, 4)
  - *Topic:* Nearest Neighbor,  $k$ -Means, and  $k$ -Medoids
  - *Readings:* Alpaydin, Chapter 7, 8
- Week 3 (September 7, 9, 11) — No class on Monday (Labor Day)
  - *Topic:* Density Estimation and Kernels
  - *Readings:* Alpaydin, Chapter 8
- Week 4 (September 14, 16, 18)
  - *Topic:* Learning Theory (PAC and Bayes)
  - *Readings:* Alpaydin, Chapter 2
  - Programming assignment 1 due: September 18
- Week 5 (September 21, 23, 25)
  - *Topic:* Decision Trees, Rule Learning, Hierarchical Clustering, Density-based Clustering
  - *Readings:* Alpaydin, Chapter 9
- Week 6 (September 28, 30, October 2)
  - *Topic:* Naïve Bayes and Variants
  - *Readings:* Alpaydin, Chapters 3, 15
- Week 7 (October 5, 7, 9)
  - *Topic:* Linear Models (Perceptrons, Logistic Regression, SVMs)
  - *Readings:* Alpaydin, Chapter 4, 10
- Week 8 (October 12, 14, 16)
  - *Topic:* Multi-Layer Networks, Autoencoders, Restricted Boltzmann Machines
  - *Readings:* Alpaydin, Chapter 11
  - Programming assignment 2 due: October 14
  - Midterm exam: October 16
- Week 9 (October 19, 21, 23)
  - *Topic:* Stacked Autoencoders, Deep Belief Networks, Convolutional Networks
  - *Readings:* Alpaydin, Chapter 12
- Week 10 (October 26, 28, 30)
  - *Topic:* Radial Basis Function Networks, Kernel Machines

- *Readings:* Alpaydin, Chapter 13, 14
- Week 11 (November 2, 4, 6)
  - *Topic:* Genetic Algorithms and Schemata
  - *Readings:* Sheppard, Chapter 4, 6
- Week 12 (November 9, 11, 13) — No class on Wednesday (Veteran’s Day)
  - *Topic:* Swarm Algorithms (ACO and PSO)
  - *Readings:* Sheppard, Chapter 5
  - Programming assignment 3 due: November 11
- Week 13 (November 16, 18, 20)
  - *Topic:* Evolution Strategies and Differential Evolution
  - *Readings:* Sheppard, Chapter 4
- Fall Break (November 23, 25, 27) — No class
- Week 14 (November 30, December 2, 4)
  - *Topic:* Advanced Topics
  - *Readings:* None
- Week 15 (December 7, 9, 11)
  - *Topic:* Advanced Topics
  - *Readings:* None
  - Programming assignment 4 due: December 11
- Final: December 14, 12:00 – 1:50